

Numerical Techniques 2026–2027

0. Welcome

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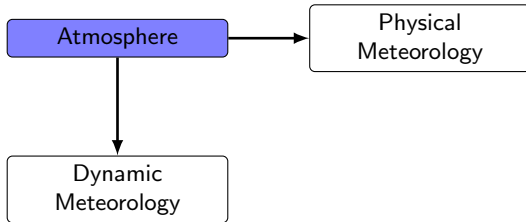
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Postgraduate Studies in Weather and Climate Modeling

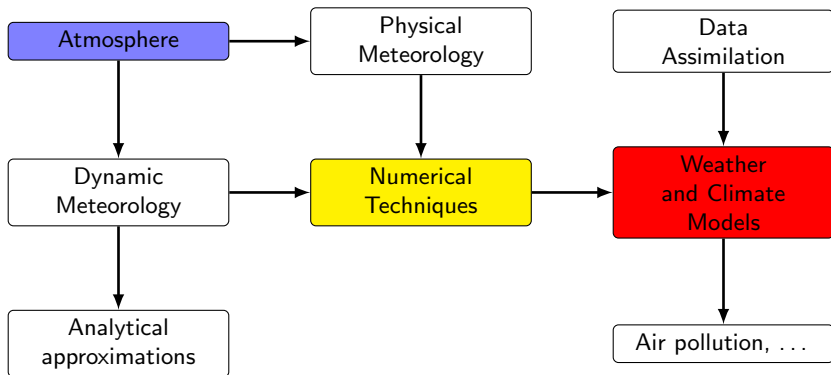
Ghent University

- Welcome
- Context: why numerical techniques
- Objectives of this course
- Course material
- Practical information

Atmosphere



Why numerical techniques?



- Get hold of problems that occur due to solving equations numerically (with a computer)

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- Don't be frightened by code
... but it's no course on programming either!

- AI-driven weather models are performing better and better, sometimes beating physics-based models on specific scores, often at a fraction of the computational cost
- *[personal assessment!]* This will make physics-based models less and less relevant, especially for operational weather forecasting
- Nevertheless,
 - ▶ Physics-based models remain (for now) important for training AI models
 - ▶ There is value in scientific understanding
 - ▶ Many aspects treated in this course are also relevant for AI models, e.g. time-stepping, stability, scalability, etc.

date	16h00	17h30
21/09	Introduction, Stability	(optional) Practicum Python basics
28/09	Time discretization	Practicum
05/10	Space discretization	
12/10	(no lesson)	
19/10	Spectral methods	Practicum
26/10	Nonlinearity	Practicum
02/11	(no lesson)	
09/11	Semi-Implicit and Semi-Lagrangian models	Project assignment
16/11	(no lesson)	
23/11	Guest lecture on AI	
30/11	Parallel computing (TBC)	Project support session
07/12	Guest lecture on a real weather model (TBC)	
14/12	Student project presentations	

- Slides will appear on Ufora
- All material (slides sources, Jupyter notebooks) available on https://github.com/ddegrauwe/ugent_numtech
- References:
 - ▶ *Numerical Methods for Wave Equations in Geophysical Fluid Dynamics*, Dale R. Durran, Springer, 1999, ISBN 0-387-98376-7.
 - ▶ *Chebyshev and Fourier Spectral Methods*, John P. Boyd, Springer, 2001, ISBN 978-3-540-51487-9.
- Some papers (depending on project)

- (Check Ufora for modifications to time schedule)
- Practical sessions
 - ▶ We will use High-Performance Computing (HPC) infrastructure of UGent: create account on
`https://www.ugent.be/hpc/en/access/faq/access`
 - ▶ access through browser via `https://login.hpc.ugent.be`
 - ▶ ... or you can just install Linux on your laptop
- Programs needed: python, Jupyter notebooks
- Evaluation: student project (2/3/4 persons) on simple model
 - ▶ presentation for other students
 - ▶ (small) report

Questions?